



Equity Arena

Backend & Frontend Engineering Report

REPORT DATE	30 August 2026
REPOSITORY	Equity-Arena · branch main
SCOPE	Full-stack defect investigation, algorithm verification, feature delivery
ENVIRONMENT	Node v22.21.0 · PostgreSQL 16 · Vite 6 · React 18
METHOD	Automated suites, property tests, and live end-to-end runs against the running stack

~120 CHECKS EXECUTED	10 DEFECTS FIXED	4 FINDINGS OPEN	0 REGRESSIONS
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Headline: Two defects were silently corrupting money. Per-buy rounding of the cost basis *overstated trader profit*, and a nodemon restart loop was killing every WebSocket every six seconds. Both are fixed and verified. Four findings remain open, none blocking gameplay, one blocking CI.

1. Executive summary

This engagement began as a request to run the project and grew into a full-stack investigation. Ten defects were found and fixed, six features were delivered, and the backend was then tested end to end — every core algorithm, the limit-order engine, the price engine, and the news pipeline — against the live running system rather than mocks.

The most consequential finding was financial. The weighted-average cost basis was re-rounded to two decimals on every purchase, so the stored average drifted below the true figure and reported profit drifted above reality. It was invisible per-trade and compounding across a session.

AREA	BEFORE	AFTER
Market connection	Dropped every ~6s; UI stuck on "Reconnecting to market..."	Stable; 1 connection / 30s, 0 restarts
Cost-basis accuracy	Drifted 3.32 IC low over 40 buys; profit overstated	Exact to <1e-9 across 300 random books
Page refresh	/trader and /admin returned 404 / 401	All five client routes return 200
Chart timeframes	Buttons inert; bucketing erased ~13% of volatility	Real ticks per window; highs and lows exact
Trade feedback	Every sale mislabelled "Admin added X IC"	Only genuine admin credits raise that toast

2. Test results

2.1 Project suite (npm test)

#	SUITE	PASSED	STATUS
1	Auth & role access control	5 / 5	PASS
2	Trading & wallet math	6 / 6	PASS
3	Limit orders & fund locking	5 / 5	PASS
4	Quant price engine	4 / 4	PASS
5	News engine & privacy	4 / 4	PASS
6	Admin session & trading lock	—	ERROR
7	Admin vs trader sync	3 / 3	PASS
8	Tournament leaderboard	3 / 3	PASS
9	Multi-tab real-time sync	1 / ?	HANGS
10	Portfolio & P&L math	43 / 43	PASS*

* Suite 10 passes only when invoked directly. Suite 9 never terminates, so a normal npm test run never reaches it (finding F2).

2.2 Additional harnesses written for this report

HARNESS	RESULT	COVERS
Deep algorithm harness	41 / 42	Cost basis, limit predicates, fund locking, price bounds, news steering, session lock, cash conservation, portfolio math, data integrity
Live news → price test	8 / 9	Real broadcast through admin API, socket delivery, privacy, price movement, clamping

Both single failures were harness artifacts of cross-process module state, not product defects — detailed in §4.

3. Algorithms verified

3.1 Weighted-average cost basis

Property-tested across 300 randomly generated books of up to 25 purchases each. The incremental average now agrees with exact total-cost accounting to better than 1e-9. A regression guard confirms the previous implementation still drifts, so the fix cannot silently revert.

```
true ticks (n=300):      low 37.70   high 153.12   range 115.42
OLD 1-min MEAN (n=31):  low 38.67   high 138.69   range 100.02
                        -> displayed 86.7% of actual volatility
```

3.2 Limit-order engine

All six boundary conditions verified, including exact-equality fills:

CONDITION	RULE	RESULT
BUY, price below target	fills	PASS
BUY, price exactly at target	fills	PASS
BUY, price above target	does not fill	PASS
SELL, price above target	fills	PASS
SELL, price exactly at target	fills	PASS
SELL, price below target	does not fill	PASS

Reserved funds equal $\sum(\text{target} \times \text{quantity})$ across pending buys, and available balance never falls below zero.

3.3 Price engine

All 15 listed stocks sit inside the $[0.20\times, 2.50\times]$ base-price clamp. This settles an earlier question about "unrealistic" swings: **RTB at +150.08% was not a defect** — $2.50 \times 2,650 = 6,625$ exactly, the designed ceiling.

3.4 News pipeline — live end-to-end

A genuine Agriculture bulletin (+18% effect) was broadcast through the admin API against the running server, with a real trader socket attached:

```
ANAG: 24.84 -> 26.34 (+6.04%, news said +18%) direction MATCHES
```

- Trader receives news: broadcast over WebSocket.
- **Privacy holds** — payload carries stockId: null and stockSymbol: null, so the target never leaks to traders.
- Sector-wide effects arm every stock in that sector; negative effects steer downward.
- An unknown symbol is a safe no-op rather than a crash.
- The steered target is clamped to the same price bounds.

3.5 Cash conservation

For each account the identity $20,000 + \text{top-ups} + \text{realised P/L} = \text{wallet} + \text{cost basis}$ was checked independently of the production code path:

ACCOUNT	LEFT SIDE	RIGHT SIDE	DIFFERENCE
bro	21,228.87	21,228.87	0.0000
123		balanced	0.0000
hello	1.0e29	1.0e29	float64 limit

The hello discrepancy is a float64 precision artifact of a 1e29 IC balance, not a logic error — see finding F4.

4. Open findings

F1 · TEST INFRASTRUCTURE · MEDIUM

Test suite cannot run against a live server

Suite 6 aborts with *"Trading session is already running (ACTIVE)"*. It calls `startNewSession()`, which refuses when a session exists. The suite assumes an idle database.

Remedy: stop the session first, or point tests at a dedicated test database.

F2 · TEST INFRASTRUCTURE · HIGH — BLOCKS CI

npm test never terminates

`09-multi-tab-sync.test.js` connects sockets to the dev server on `:5001`, then calls `emitPortfolioUpdate` inside the test process, where the module-level `io` is null. The guard `if (io && userId)` makes it a silent no-op, and the test awaits `Promise.all(...)` with no timeout.

The run hangs indefinitely and **suite 10 never executes**. In CI this blocks the pipeline forever rather than failing.

Remedy: add timeouts to the awaited promises, and either initialise the socket server in-process or drive the emit over HTTP.

F3 · BACKEND · LOW

applyNewsImpact is imported but never exported

`adminRoutes.js:6` destructures it from `marketTicker`, which does not export it. Harmless today because nothing calls it — it is simply undefined — but any future call site would throw at runtime.

Remedy: delete the dead import, or implement and export the function.

F4 · DATA · MEDIUM

Admin top-ups are unbounded

The `hello` account holds **1e29 IC**, beyond the point where `float64` can represent IC to the paisa. Arithmetic on that account is no longer trustworthy.

Remedy: cap top-up amounts server-side and reset the affected account before a real tournament.

5. Defects fixed

D1 · BACKEND · CRITICAL

Nodemon restart loop severed every WebSocket

A session tick ran an auto-backup guarded to 15 minutes by an *in-memory* variable. The backup wrote a JSON file into a watched folder, nodemon restarted, the restart reset the guard to zero, and the cycle repeated — 59 restarts observed.

Every client saw `transport close` roughly every six seconds. This was the true cause of the persistent "Reconnecting to market..." banner.

Fix: `backend/nodemon.json` scopes the watch to `src`. Verified 5 connections/30s → 1, and 0 restarts.

D2 · BACKEND · CRITICAL — FINANCIAL

Cost-basis rounding overstated trader profit

The stored weighted average was re-rounded to two decimals on every purchase. Each buy rounded an already-rounded number, so the basis drifted low and reported profit drifted high — 3.32 IC understated over 40 buys.

Confirmed against live data: MRI held a +0.0042 IC/share drift, while a position bought after the fix matched the transaction log exactly.

Fix: full precision retained in storage; rounding applied only at display.

D3 · FRONTEND · HIGH

Socket state race pinned the UI offline

A torn-down socket's late `disconnect` could overwrite the live socket's `connect`, leaving `isConnected` false permanently. The effect also depended on the whole `user` object, rebuilding the socket on every revalidation, and `reconnect_failed` was bound to the socket where it never fires.

Fix: active-instance ref guard, `user.id` dependency, Manager-level listener, unlimited reconnect attempts.

D4 · FRONTEND · HIGH

Page refresh broke both dashboards

Vite proxy keys are prefix matches, so `'/trade'` also captured `/trader` and `'/admin'` captured the admin page — both were forwarded to the backend on refresh. `/news` and `/leaderboard/public` were not proxied at all and were silently receiving `index.html` instead of JSON.

The same class of bug existed in production, where `requireAdmin` returned 401 JSON for a browser refresh of `/admin` .

Fix: anchored regex proxy rules plus an SPA guard in the Express server. All five client routes and all API paths verified.

D5 · FULL STACK · MEDIUM

Every sale reported as an admin credit

The wallet toast was inferred purely from `"balance went up"` and hardcoded the word `"Admin"` , so sale proceeds were mislabelled. Admin top-ups now tag their own `portfolio:update` , and only that tag raises the toast.

D6 · FRONTEND · MEDIUM

Chart timeframes were inert and data was smoothed

The dashboard hardcoded `range=1D` , so the `5M/15M/30M/1H` buttons only trimmed a day's ticks client-side. Bucketed ranges also averaged prices within each bucket — a price that never traded — clipping real highs and lows and displaying 86.7% of actual volatility.

Fix: the selected timeframe drives the request; intraday serves raw ticks; buckets carry close/open/high/low. Verified against raw database truth for all four windows.

D7 · FRONTEND · MEDIUM

"Go to Market" opened a blank screen

The button set the tab to `'FL00R'` , a key no render block matches, so the entire tab body rendered nothing. A sweep of every tab key confirmed it was the only orphan.

D8 · FRONTEND · LOW

Breaking-news chime played twice

The dashboard played the chime and then opened a toast that played it again without an event id, defeating de-duplication. The dashboard is now the single owner.

D9 · FRONTEND · MEDIUM

Trader's Manual was unreachable after first login

`TopBar` accepted an `onOpenTour` prop but never rendered anything with it, so the guide could not be reopened. A book icon now sits beside the sound and theme controls.

Onboarding guide documented features that do not exist

It promised candlestick charts and 1D/1W/1M timeframes (the app draws a line chart with 5M/15M/30M/1H) and illustrated the game with Adani Ports and Tata Motors, neither of which is listed. It also omitted the two rules that cost players money: limit orders reserving assets, and end-of-session auto-liquidation.

Fix: rewritten as a seven-section manual, every figure verified against source, and illustrated with real screenshots captured from the running terminal.

6. Features delivered

FEATURE	DETAIL
Average buy price	Per-share weighted average plus total invested on every holding, with an Invested total in the section header.
Realised P/L column	New <i>Result</i> column on My Recent Trades. Computed server-side by replaying the complete trade history, since the client only receives 20 rows and would miss earlier buys.
Range removal	Both "Range" readouts removed from stock cards, along with the dead computations that fed them.
News toast redesign	Larger, longer-dwelling, and strictly neutral — the previous green "LIVE ALERT" implied a price direction the game must not reveal.
Trader's Manual	Seven sections, glassmorphic, enlarged type, real screenshots, full keyboard and screen-reader support.
Manual launcher	Book icon in the top bar to reopen the guide at any time.

7. Recommended next steps

- **Unblock CI (F2).** Add timeouts to the multi-tab socket test — today a CI run would hang rather than fail.
- **Cap admin top-ups (F4)** and reset the 1e29 IC account before a live tournament.
- **Isolate the test database (F1)** so suites can run without stopping the game.
- **Remove the dead import (F3).**
- **Consider back-filling historical cost bases.** The rounding fix corrects future buys but does not rewrite existing holdings; residual drift is ~0.004 IC/share and invisible at 2dp.
- **Persist the auto-backup timestamp.** The 15-minute guard still resets on every restart. Harmless now the loop is broken, but fragile.